

Improving Transformer-Based Underwater Semantic Segmentation Through Domain-Specific Fine-Tuning

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Abstract. Object detection and pixel-level segmentation in underwater imagery play a key role in marine conservation and underwater robotic activities. However, underwater environments present major challenges for computer vision; scattered light, color distortion, and poor visibility all reduce segmentation accuracy. Because of these effects, models trained on terrestrial datasets often struggle to handle underwater images. The performance of Transformer-based architectures degrades under underwater conditions although it offers powerful feature extraction capabilities. This study addresses this limitation through fine-tuning SegFormer and Mask2Former transformer models. We then assess their performance before and after fine-tuning to determine the effectiveness of domain-specific adaptation in complex scenes featuring diverse object categories and varying lighting conditions. We observed significant improvements across all the metrics examined. Both models demonstrate better region localization and improved semantic consistency after adaptation to underwater data. The findings highlight that model adaptation to the target domain plays a significant role in understanding the underwater scene and can significantly improve the reliability of transformer-based segmentation models for real-world underwater applications.

Keywords: Object detection · pixel-level segmentation · computer vision · marine conservation · semantic consistency

1 Introduction

Recent advances in underwater imaging have created new opportunities in marine research, autonomous underwater navigation, environmental monitoring, seabed inspection, and underwater robotics. Many of these applications rely on accurate scene understanding, where identifying and separating different objects present in an underwater image becomes essential. Semantic segmentation plays an important role in this process by assigning a class label to every pixel in the image, allowing detailed interpretation of underwater scenes. Despite significant progress in semantic segmentation for natural images, underwater image segmentation remains a challenging task. Understanding underwater scenes remains a

challenging problem in computer vision because underwater images are often degraded by poor visibility, color distortion, scattering effects, uneven lighting, and blurred object boundaries. In addition, complex backgrounds and low contrast between objects make accurate segmentation more difficult. These challenges often reduce the quality of feature extraction and lead to poor segmentation results. Consequently, segmentation models trained on conventional benchmark datasets often fail to generalize well when applied directly to underwater imagery.

Transformer-based [1] models have recently become one of the most effective approaches for semantic segmentation and other computer vision tasks. Among these, SegFormer [2] and Mask2Former [3] have achieved strong results by learning detailed contextual information and capturing relationships between different regions of an image. Despite their success, these models are mainly trained using large-scale datasets containing images captured in terrestrial environments. Due to this, they do not perform very well in underwater imaging applications.

Processing underwater imaging is challenging. As light moves through water, it gets absorbed and scattered, which washes out colors and makes it hard to see clearly. It may also end up with uneven lighting and blurry edges around objects. Moreover, floating particles in the water blur out fine details. Most segmentation models are trained on regular land-based images and struggle on underwater scenes. Fine-tuning these models on underwater data helps them adjust to these conditions while still holding onto what they learned during their original training.

This study looks at how domain-specific fine-tuning affects transformer-based semantic segmentation models. SegFormer [2] and Mask2Former [3] are fine-tuned on the SUIM [4] dataset, with their segmentation performance measured both before and after adaptation. This comparison sheds light on how fine-tuning boosts model robustness and segmentation accuracy in challenging underwater environments.

The major contributions of this research are as follows:

- Evaluate the baseline segmentation capability of SegFormer and Mask2Former on underwater imagery.
- Adapt both models through fine-tuning using the SUIM dataset.
- Measure the improvement achieved through domain adaptation using IoU, Dice Score, and Pixel Accuracy.
- Compare pretrained and fine-tuned models to analyze the effectiveness of fine-tuning for underwater semantic segmentation.

2 Related Work

Semantic segmentation is a fundamental task in computer vision. It involves assigning a category label to every pixel in an image. This pixel-level precision makes it useful in a wide range of applications. Such applications include autonomous systems, medical image analysis, environmental monitoring and underwater robotics. Semantic segmentation approaches relied primarily on Fully

Convolutional Networks (FCNs) [5]. It demonstrated strong dense prediction capability and established the foundation for later encoder-decoder architectures.

U-Net [6] improved segmentation accuracy by introducing skip connections between the encoder and the decoder. It helps in recovering fine spatial details that were typically lost during downsampling. DeepLab [7] took a different approach, combining dilated convolutions with spatial pyramid pooling to capture contextual information across multiple scales, which allowed it to perform better in more visually complex scenes.

While CNN-based methods achieved strong results, they had difficulty modeling long-range dependencies because convolution operations are inherently local in scope. Transformer architectures [1] offered a compelling solution to this limitation. The Vision Transformer [8] showed that self-attention can effectively capture global relationships across an image. SegFormer [2] built on this idea with a hierarchical transformer encoder paired with a lightweight MLP decoder, striking a practical balance between accuracy and computational cost. Its design also preserves multi-scale feature representations without depending on positional encoding, which makes it well suited for dense prediction tasks.

Mask2Former [3] extended transformer-based segmentation by introducing masked attention together with a unified mask classification framework capable of handling semantic, instance, and panoptic segmentation within a single architecture. The model achieved state-of-the-art results on several benchmark datasets and demonstrated strong robustness across a wide variety of challenging visual scenes.

For underwater applications, semantic segmentation remains challenging because of poor illumination, scattering, color distortion, and low contrast. To address the lack of benchmark datasets in this domain, Islam *et al.* [4] introduced the SUIM dataset, which contains pixel-level annotations for diverse underwater objects including fish, reefs, plants, robots, divers, and seafloor regions. This dataset has become one of the most widely used benchmarks for underwater scene understanding.

Several studies have since focused on improving segmentation specifically for underwater imagery. Li *et al.* [9] explored underwater image enhancement before segmentation to improve visibility and feature extraction. Liu and Fang [10] combined image correction with DeepLab-based segmentation to improve performance under severe color degradation. Wang *et al.* [11] proposed WaterBiSeg-Net to improve boundary-aware segmentation of marine targets through multi-scale feature enhancement.

More recently, transformer-based underwater segmentation methods have shown promising results. Improved SegFormer variants using Swin Transformer [12] backbones and attention enhancement modules have reported better segmentation accuracy on SUIM by improving global feature extraction and multi-scale representation. Similarly, UWSegFormer [13] introduced underwater image quality attention and multi-scale aggregation mechanisms to enhance boundary preservation and segmentation completeness in challenging underwater environments.

Despite these advances, accurate segmentation of underwater imagery remains difficult because of object-scale variation, blurred boundaries, low visibility, and environmental noise. These challenges have driven the development of more resilient transformer-based architectures that integrate global contextual information with efficient feature fusion to achieve more accurate underwater semantic segmentation.

3 Transformer-Based Semantic Segmentation Models

For underwater semantic segmentation, two transformer-based architectures were investigated: SegFormer [2] and Mask2Former [3]. Both models were adapted to the SUIM [4] dataset with eight semantic categories.

3.1 SegFormer-Based Segmentation

The SegFormer model is a hierarchical transformer-based architecture consisting of a multi-scale transformer encoder followed by a lightweight multilayer perceptron (MLP) decoder. This architectural design allows efficient extraction of both local and global contextual representations without relying on computationally expensive attention decoding mechanisms.

In this work, a pretrained SegFormer-B2 model was utilized, originally learned from a large-scale scene understanding dataset. The architecture was adapted for underwater semantic segmentation by modifying the final classification layer to predict eight output classes corresponding to the SUIM label space. Formally, Let the feature representation generated by the encoder be represented as $F \in \mathbb{R}^{H \times W \times C}$. The segmentation head learns a mapping:

$$\hat{Y} = f_{\theta}(F) \quad (1)$$

where $\hat{Y}$ represents the pixel-wise class probability distribution over the eight classes. During fine-tuning, the encoder parameters were initialized from pre-trained weights, while the segmentation head was randomly initialized and optimized for underwater domain adaptation.

3.2 Mask2Former-Based Segmentation

The Mask2Former model is a unified transformer architecture built to address semantic, instance, and panoptic segmentation simultaneously. The model utilizes a masked attention module mechanism that learns a fixed number of query embeddings, each responsible for predicting a segmentation mask and its corresponding class label.

In this study, a pretrained Mask2Former model with a Swin-Small backbone was used. The classification layer was modified to output eight semantic classes for the SUIM dataset. Given a set of learned queries $\{q_i\}_{i=1}^N$, the model predicts a set of masks $\{m_i\}$ and class probabilities $\{c_i\}$:

$$(m_i, c_i) = g_{\phi}(q_i, F) \quad (2)$$

where F denotes the encoded image features and g_ϕ represents the transformer decoder. The final segmentation map is obtained by aggregating all predicted masks through a per-pixel argmax operation.

4 Methodology

4.1 Dataset Preparation

The SUIM [4] dataset comprises more than 1,500 images accompanied by pixel-wise annotations spanning eight object categories: fish (vertebrates), reefs (invertebrates), aquatic plants, wrecks/ruins, human divers, robots, and sea-floor. The dataset was assembled from underwater exploration activities and human-robot collaborative studies, followed by manual annotation performed by trained annotators.

Training was performed using RGB underwater images alongside their pixel-level segmentation masks. Both the images and masks were resized to 256×256 before being processed by the model. Image resizing ensured a consistent spatial resolution across the dataset while reducing computational overhead during training. Images were separated into train, test, and valid sets with a ratio of 70:15:15.

The segmentation masks in SUIM are provided as color-coded annotations. To enable supervised training, each RGB mask was converted into a class-index mask where each pixel corresponds to one of the predefined semantic labels. This conversion was performed by mapping the RGB channels into integer label values in the range 0–7, resulting in a total of eight semantic classes.

4.2 Training Procedure

For the SegFormer model, the optimization is performed using the standard pixel-wise cross-entropy loss. For an input image of spatial size $H \times W$, given the predicted segmentation map $\hat{Y}$ and true output labels Y , the loss is defined as:

$$\mathcal{L}_{CE} = -\frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W \sum_{c=1}^C y_{ijc} \log(p_{ijc}) \quad (3)$$

Here, H and W are the height and width of the segmentation map, respectively, and C is the total number of semantic classes. The term y_{ijc} refers to the ground-truth label for class c at pixel location (i, j) , and p_{ijc} is the corresponding predicted probability.

For Mask2Former, a composite loss function is used, combining binary cross-entropy loss and Dice loss for mask supervision, along with a classification loss for predicted object queries.

The mask loss is defined as:

$$\mathcal{L}_{\text{mask}} = \lambda_{ce} \mathcal{L}_{bce} + \lambda_{dice} \mathcal{L}_{dice} \quad (4)$$

where $\mathcal{L}_{bce}$ denotes the binary cross-entropy loss, whereas $\mathcal{L}_{dice}$ represents the Dice loss. The weighting factors are set to:

$$\lambda_{ce} = 5.0, \quad \lambda_{dice} = 5.0.$$

The final loss function is given by:

$$\mathcal{L}_{\text{Mask2Former}} = \mathcal{L}_{\text{mask}} + \lambda_{cls} \mathcal{L}_{cls}, \quad (5)$$

where $\mathcal{L}_{cls}$ denotes the classification loss for matched queries and $\lambda_{cls} = 2.0$.

The matching between predicted masks and ground truth segments is performed using bipartite matching, ensuring optimal assignment between predicted queries and ground truth labels.

To summarize, SegFormer relies on a simple cross-entropy loss for dense pixel classification, whereas Mask2Former employs a multi-component loss combining mask-level supervision (BCE + Dice) and query-based classification loss.

4.3 Training Strategy

To adapt the pretrained models to underwater imagery, domain-specific fine-tuning was performed on the SUIM training set. Instead of training the network from scratch, pretrained weights were used to initialize the network as part of a transfer learning approach and updating the parameters using underwater data. This allows the model to preserve general visual representations learned from large-scale datasets while adapting them to underwater-specific visual characteristics.

Input images were processed using the corresponding models image processor, which applies normalization and tensor conversion compatible with the pre-trained backbone. During training, the processed image tensors were paired with their corresponding semantic masks and provided as input to the network.

The model outputs a pixel-wise segmentation prediction map. Since the output resolution of the model differs from the mask resolution, the predicted logits were resized to the ground-truth mask size using bilinear interpolation before computing the training loss.

5 Experimental Results

The network was trained using the AdamW optimizer with an initial learning rate of 6×10^{-5} . A batch size of 8 was used for training, and both the models were fine-tuned for 10 epochs.

The total loss was computed over each mini-batch and backpropagated through the network. Model parameters were updated iteratively after each batch using gradient descent optimization. The complete set of training parameters for SegFormer is summarized in Table 1 and for Mask2Former is provided in Table 2.

Training was performed using GPU acceleration when available to improve computational efficiency.

Table 1: Training configuration used for fine-tuning SegFormer [2] on the SUIM [4] dataset

Parameter	Value
Model Backbone	SegFormer-B2
Pretrained Weights	ADE20K pretrained
Input Image Size	256×256
Number of Classes	8
Batch Size	8
Learning Rate	6×10^{-5}
Optimizer	AdamW
Number of Epochs	10
Hardware	GPU (CUDA-enabled)

Table 2: Training configuration used for fine-tuning Mask2Former on the SUIM dataset

Parameter	Value
Model Backbone	Swin-Small Transformer
Input Image Size	256×256
Number of Classes	8
Batch Size	8
Learning Rate	1×10^{-5}
Optimizer	AdamW
Number of Epochs	10
Hardware	GPU (CUDA-enabled)

Table 3: Computational comparison of evaluated segmentation models

Model	Trainable Parameters	Model Size	Inference Time	GPU Memory
SegFormer	27,352,776	109.41 MB	0.9 s	2.1 GB
Mask2Former	68,721,043	265.01 MB	0.9 s	1.8 GB

Table 4: Performance comparison of SegFormer and Mask2Former before and after fine-tuning on the SUIM dataset

Model	IoU	Dice Score	Pixel Accuracy	Training Time
SegFormer (before fine-tuning)	0.0362	0.0580	0.1854	–
SegFormer (after fine-tuning)	0.6163	0.7023	0.8646	350 min
Mask2Former (before fine-tuning)	0.0236	0.0398	0.0747	–
Mask2Former (after fine-tuning)	0.6326	0.7110	0.8808	70 min

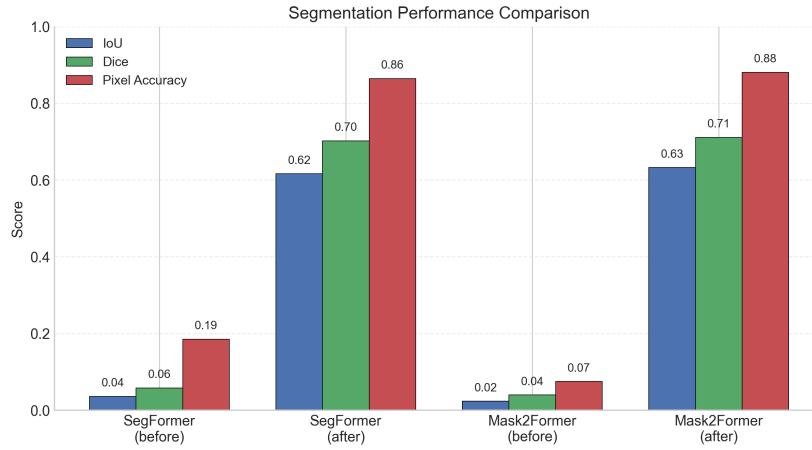


Fig. 1: Segmentation performance comparison of different models before and after training. Metrics include IoU, Dice score, and pixel accuracy.

After completion of training, the fine-tuned models weights along with the associated preprocessing configuration were saved for later evaluation and inference. Saving both the trained network and preprocessing pipeline ensures reproducibility and allows direct deployment of the models for underwater scene segmentation on unseen images.

The resulting fine-tuned models were then used for performance evaluation and comparison against the baseline pretrained SegFormer and Mask2Former model to analyze the effect of domain adaptation on underwater semantic segmentation accuracy.

Table 4 and Figure 1 present the performance of SegFormer and Mask2Former before and after fine-tuning on the SUIM dataset. The illustrative examples are shown in Figure 2 and Figure 3. Before fine-tuning, both models achieved very poor segmentation performance, with IoU values of 0.0362 for SegFormer and 0.0236 for Mask2Former. These low scores indicate that the pretrained models were not well adapted to the underwater domain. Similarly, both methods produced Dice scores below 0.06, reflecting minimal agreement between the predicted segmentation masks and the ground-truth annotations. Fine-tuning substantially improved the performance of both models. SegFormer achieved an IoU of 0.6163 and a Dice score of 0.7023, whereas Mask2Former obtained slightly higher values of 0.6326 and 0.7110, respectively. A comparable improvement in pixel accuracy was observed, increasing to 0.8646 for SegFormer and 0.8808 for Mask2Former. Training efficiency also varied noticeably between the two models. Mask2Former reached convergence in around 70 minutes, whereas SegFormer took nearly 350 minutes to do the same. This shows that Mask2Former is far more computationally efficient option for this task. The computational comparison of the evaluated segmentation models is shown in Table 3.

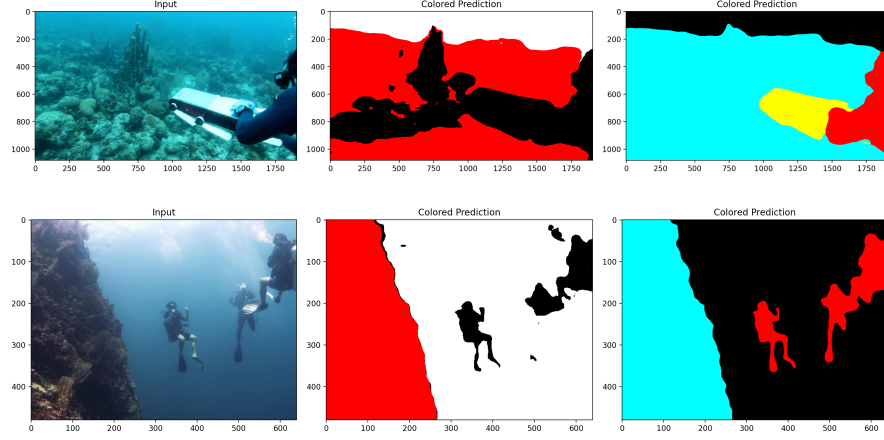


Fig. 2: Representative segmentation results on the SUIM dataset. From left to right, the images show the original underwater scene, the segmentation prediction generated by the pretrained Mask2Former model before and after fine-tuning.

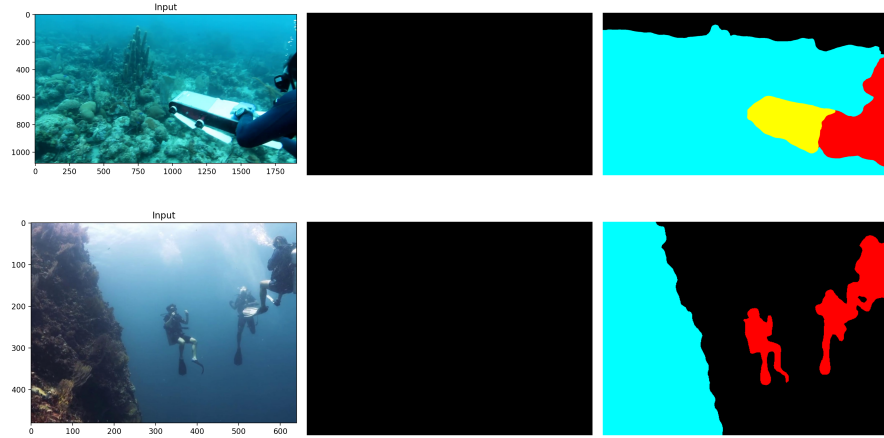


Fig. 3: Representative segmentation examples from the SUIM dataset. The columns illustrate the input image, the segmentation mask predicted by the pretrained SegFormer model prior to fine-tuning, and the refined prediction after fine-tuning.

The code for this work is publicly available at: <https://github.com/SA-32/Improving-Transformer-Based-Underwater-Semantic-Segmentation-Through-Domain-Specific-Fine-Tuning>

6 Conclusion

Accurate semantic segmentation in underwater images remains a challenge, largely due to color degradation and poor visibility of key features. In this work, we fine-tuned two transformer-based segmentation models using the SUIM dataset. Experimental analysis shows that fine-tuning significantly improves segmentation quality across multiple performance measures such as Intersection over Union (IoU), Dice coefficient, and pixel-wise accuracy. The results also demonstrate that adapting pretrained transformer models to underwater data helps reduce the performance gap caused by domain shift and improves semantic consistency across multiple object categories. In this work, we achieved significant improvements in the performance of the transformer-based models and concluded that domain-level fine-tuning significantly improves the performance. Future work includes extending this approach to datasets from multiple domains and evaluating it with various foundation models to analyze performance and explore possible improvements.

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